High-Level Requirements Document (HLR)

Hospital Patient Registration Management System

1. Project Overview

The Hospital Patient Registration Management System is designed to streamline the patient registration process by capturing patient demographic information, contact details, and emergency contact information. Upon successful registration, the system will generate an invoice for the applicable registration charges.

The system will maintain accurate patient records and provide authorized hospital staff with the ability to register and retrieve patient information.

2. Project Objectives

- Simplify and streamline patient registration.
- Maintain accurate and centralized patient demographic records.
- Capture patient contact and emergency contact information.
- Generate a unique patient identification number.
- Automatically generate an invoice after successful registration.
- Improve accuracy and reduce manual data entry errors.

3. Modules Required

Module 1: Patient Registration

Description: Allows authorized hospital staff to register new patients by entering their personal and demographic information.

High-Level Requirements:

- The system shall allow staff to register new patients.
- The system shall capture patient demographic information.
- The system shall generate a unique Patient ID upon successful registration.
- The system shall validate mandatory fields before completing registration.
- The system shall prevent duplicate patient registration based on configured matching criteria.
- The system shall save patient information securely.
- The system shall allow authorized staff to retrieve existing patient records.

Module 2: Patient Demographic Information

Description: Captures the patient's basic personal and demographic details.

Information to be captured:

- First Name
- Middle Name (Optional)
- Last Name
- Date of Birth
- Age (System-calculated)
- Gender
- Marital Status
- Blood Group (Optional)
- Nationality
- Identification Type
- Identification Number

High-Level Requirements:

- The system shall allow staff to enter patient demographic information.
- The system shall validate mandatory demographic fields.
- The system shall calculate patient age based on Date of Birth.
- The system shall validate the date of birth and identification information.
- The system shall store demographic information against the patient's unique ID.

Module 3: Patient Contact Information

Description: Captures the patient's primary contact information and residential address.

Information to be captured:

- Primary Phone Number
- Alternate Phone Number (Optional)
- Email Address (Optional)
- Address Line 1
- Address Line 2 (Optional)
- City
- State
- ZIP/Postal Code
- Country

High-Level Requirements:

- The system shall allow staff to enter patient contact details.
- The system shall validate phone number and email formats.
- The system shall capture the patient's residential address.
- The system shall allow optional alternate contact information.

- The system shall associate contact information with the patient's registration record.

Module 4: Emergency Contact Information

Description: Captures emergency contact details for communication in case of an emergency.

Information to be captured:

- Emergency Contact Full Name
- Relationship to Patient
- Primary Phone Number
- Alternate Phone Number (Optional)
- Email Address (Optional)
- Address (Optional)

High-Level Requirements:

- The system shall allow staff to enter emergency contact information.
- The system shall capture the emergency contact's relationship with the patient.
- The system shall validate the emergency contact phone number.
- The system shall allow staff to save emergency contact information as part of patient registration.
- The system shall allow authorized staff to view emergency contact details.

Module 5: Registration Fee and Invoice Generation

Description: Automatically generates an invoice after successful patient registration.

High-Level Requirements:

- The system shall calculate the applicable patient registration fee.
- The system shall generate an invoice after successful registration.
- The system shall assign a unique invoice number.
- The invoice shall include patient information and registration details.
- The invoice shall display the registration fee, applicable taxes, discounts (if any), and total payable amount.
- The system shall include invoice generation date and payment due date, if applicable.
- The system shall allow staff to view and print the generated invoice.
- The system shall maintain invoice history associated with the patient record.

Invoice Information:

- Invoice Number
- Invoice Date
- Patient ID
- Patient Name

- Registration Date
- Description of Charges
- Registration Fee
- Tax (if applicable)
- Discount (if applicable)
- Total Amount
- Payment Status (Paid / Unpaid / Partially Paid)

4. Overall System Workflow

1. Hospital staff accesses the patient registration system.
2. Staff selects the New Patient Registration option.
3. Staff enters patient demographic information.
4. Staff enters patient contact information.
5. Staff enters emergency contact information.
6. Staff reviews the entered information.
7. Staff submits the registration.
8. System validates all mandatory information.
9. System checks for potential duplicate patient records.
10. System generates a unique Patient ID.
11. System saves the patient registration details.
12. System calculates the applicable registration fee.
13. System automatically generates an invoice.
14. Staff views, prints, or provides the invoice to the patient.

5. Non-Functional Requirements

- **Security:** The system shall ensure patient information is accessible only to authorized users.
- **Data Privacy:** The system shall protect sensitive patient information in accordance with applicable healthcare privacy regulations.
- **Performance:** Patient registration and invoice generation should be completed within an acceptable response time.
- **Availability:** The system should be available during hospital operating hours.
- **Usability:** The interface should be simple and user-friendly for hospital registration staff.
- **Data Integrity:** The system shall ensure patient registration and invoice information are stored accurately.
- **Audit Trail:** The system should maintain logs of patient registration and modification activities.
- **Scalability:** The system should support increasing numbers of patients and registrations.

6. Assumptions and Dependencies

- Hospital staff will have authorized login credentials.
- Registration charges will be configured in the system.
- Patient ID and invoice numbers will be generated automatically.
- Registration fees may vary based on hospital-defined pricing rules.
- Payment processing is outside the initial scope unless integrated separately.
- The system will support the applicable healthcare data privacy requirements.

7. Out of Scope

The following modules are not included in the initial release:

- Appointment Scheduling
- Doctor Management
- Electronic Medical Records (EMR)
- Laboratory Management
- Pharmacy Management
- Insurance Claims Processing
- Inpatient Admission and Discharge
- Billing for consultations, procedures, and treatments
- Payment Gateway Integration

8. High-Level Success Criteria

The system will be considered successful when:

- Hospital staff can successfully register a new patient.
- All required demographic and contact information is captured.
- Emergency contact information is saved successfully.
- A unique Patient ID is generated for each new patient.
- Duplicate patient registrations are identified based on configured rules.
- An invoice is automatically generated after successful registration.
- Staff can view and print the invoice.
- Patient registration and invoice records can be retrieved accurately.

Test case are here- Patient Registration Test case